



Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

PhD project: Soil Respiration Partitioning with Automated Root Measurements

We invite applications for fully-funded (candidates eligible for EU and UK fees only) 4-year PhD project within the Plant-Soil Function group in the Discipline of Botany, School of Natural Sciences at Trinity College Dublin.

Project background and description

Soil respiration is the second biggest carbon flux in the terrestrial biosphere and but the atmosphere but intrinsically difficult understand and predict because it is made up of contributions from various sources such as plants and soil microbes.

In this position were seeking a candidate interested in soil ecology, root traits and fieldwork and linking this with automated 'minirhizotron systems able to make extremely high time frequency images of roots *in situ* in the field (<https://doi.org/10.1093/jxb/erac427>). The entire, multi-person project will interpret the dynamics of root properties in a field site and use these in a new way to dynamically 'partition' soil respiration between the activities of the organisms that contribute to it at scales from seasonal to sub-daily. The results will contribute to our ability to understand and predict this critical component of ecosystem function.

You will work within group spanning ecological science, engineering, and computer science methods and specifically develop aspects of the project around 'ground truth' measurements of root properties and carbon flow, potentially also including isotope experiments and/or characterization of soil microbial communities. You will be part of a team which will develop instruments and analysis, upscale, and incorporate this information into modelling frameworks.

Fieldwork on this project will take place in Ireland but there will also be the potential to use data from sites in the UK and Spain. Root and soil analysis facilities, as well as computing resources are available within the wider research group. International collaborations, conference attendance and research trips are also expected as part of this project.

Candidate profile

The ideal candidate will:

- Hold a bachelor's or master's degree in ecology, geosciences, environmental science, plant sciences, or another relevant discipline.

- Have experience in root, plant-soil, or soil science, preferably under field conditions, or demonstrable related experience in an academic or non-academic context and willingness to engage
- Have capacity/willingness to think critically and creatively, and develop into a researcher in control of their own hypothesis-driven project
- Have demonstrable quantitative skills, including ecologically-relevant statistical methods
- Have experience in interdisciplinary projects or similar settings and a willingness to engage cross discipline.
- Meet the Trinity College Dublin [postgraduate entry requirements](#)
- Preferably hold a valid driving license
- Be able to start a PhD in September 2025

Funding

This is a 4-year PhD project and covers an annual tax-free stipend of €25,000, as well as project costs and EU student fees (funder constraint). EU fees apply to anyone who has worked or studied in the EU or UK for at least 3 out of the last 5 years, more details here: <https://www.tcd.ie/academicregistry/fees-and-payments/applicants/what-are-my-fees/>.

Application

Please send a CV and a 1-page personal statement detailing why you are interested in the project and names and contact details of two referees, with first review on the 6th June 2025 17:00 GMT by email to Dr. Richard Nair richard.nair@tcd.ie with the subject line '2025 Soil Respiration PhD application'

Given sufficient suitable applications, interviews for this position will be held in June 2025. Interviews will be held online.

The personal statement should contain details of why you want to do a PhD and why you want to do this particular PhD.

We strive for a bias free recruitment process, so we ask you to not send CVs that include a photo or information of a personal nature (e.g. age, marital status, nationality). Statements will be read before CVs. We encourage applications from underrepresented groups in STEM.

Please address all enquiries by email to Dr. Richard Nair, richard.nair@tcd.ie

Links

Botany at Trinity College Dublin: <https://www.tcd.ie/Botany/>

Plant Soil Function Group: <https://richardnair.github.io/>